Non-Markovian Dynamics in Computational Biology

Hidden-Sector Memory and the Methodological Limits of Single-Cell Analysis

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Abstract

The single-cell bioinformatics literature has documented a series of methodological puzzles over the past five years: RNA velocity methods that reverse trajectories on validated developmental data; gene regulatory network inference accuracy that "marginally exceeds random predictions" despite extensive method development; deep-learning foundation models (scGPT, Geneformer, scFoundation) that systematically underperform simple linear baselines on perturbation prediction; multi-omics integration methods that produce inconsistent results across modalities. Each puzzle has been treated as a methodological problem to be solved with better algorithms, more data, or different model architectures.

This paper applies the Observational Incompleteness (OI) framework [1] to these puzzles. The framework's C1–C4 conditions — non-trivial coupling (C1), memory persistence (C2), high-capacity hidden sector (C3), and history read-back (C4) — produce, in the memory-bearing sector they characterize, non-Markovian dynamics (P-indivisibility the stronger diagnostic, established at recurrence by C1 and finiteness alone) whenever a fast visible sector's hidden coupling realizes them. The companion Medicine paper [2] establishes that chromatin layers satisfy C1–C4 in biological systems, with DNA methylation, histone modifications, and chromatin state functioning as the slow hidden sector with timescales spanning seconds (chromatin binding) to generations (DNA methylation maintenance). This paper shows that the same structural conditions explain the bioinformatics field's documented failure modes: each puzzle has the same underlying cause — methods assume Markovian dynamics on systems with substantial hidden-sector memory, and this assumption violation produces characteristic systematic errors.

We develop the analysis across five domains where the framework's predictions match documented empirical failure modes: trajectory inference (RNA velocity assumptions vs. multi-omic velocity improvements); gene regulatory network inference (correlation-based methods outperforming causal methods, multi-omic methods outperforming RNA-only); perturbation prediction (deep-learning foundation models underperforming linear baselines, knowledge-graph methods improving on data-driven methods); multi-omics integration (static factor analysis vs. emerging timescale-aware methods); and cancer methylome clas-

sification (95% subclass accuracy of methylation-based tumor classifiers reflecting substratum-preservation in cancer). We also extend the framework's reader-writer disorder class [Medicine §10.7] into bioinformatic predictions, including a testable claim about variant pathogenicity stratification for reader/writer/eraser genes.

The structural content of the framework's application to computational biology is informational rather than algorithmic. Standard methodology assumes that more data and more complex models will overcome any prediction task; the framework identifies an information-theoretic ceiling that no transcriptome-only model can exceed, no matter how scaled. The methodological roadmap forward is to include hidden-sector information explicitly — through multi-modal data (chromatin accessibility, histone marks, methylation), knowledge graphs (encoding hidden regulatory structure), or explicit memory state representations — rather than to scale transcriptome-only architectures further.

Status — conjectural extension. This paper applies the core framework to bioinformatics and evolutionary biology, and is offered as a *conjecture*, not an established result: the account is explicitly retrodictive — the phenomena predate the framework — and its principal forward discriminator (the Hurst-exponent signature) does not by itself separate the framework from a mundane superposition null, so the conjecture awaits a test that distinguishes it from that baseline. It carries no evidential weight for the core framework — which stands or falls independently on its physics — and should be read as a structural possibility to be tested, not as support for the framework.

1. Introduction

1.1 The bioinformatics field's persistent puzzles

Single-cell transcriptomics has revolutionized biology over the past decade, producing datasets of millions of cells with detailed gene expression profiles. The computational methodology has evolved rapidly: dimensionality reduction (UMAP, t-SNE), clustering (Leiden, Louvain), trajectory inference (Monocle, Slingshot, scVelo), gene regulatory network inference (SCENIC, GENIE3), perturbation modeling (CPA, GEARS, scGPT), and multi-omics integration (MOFA, Seurat, LIGER). Each new method generation has promised improved accuracy and broader applicability, and the field's expected trajectory has been continuous improvement.

A different pattern has emerged. Recent benchmarking studies have documented systematic failures across multiple method classes:

- **RNA velocity** methods (velocyto, scVelo) "fail to identify, and even reverses the trajectory" on validated developmental datasets [3]. The failures are reproducible across at least six independent datasets [3].

- **Gene regulatory network inference** accuracy "marginally exceeds random predictions" despite a decade of method development [4]. Causal methods consistently fail to outperform correlation-based baselines — a "persistent puzzle which casts doubt on the value of causality for this task" [5].
- **Deep-learning foundation models** for single-cell biology (scGPT, Geneformer, scFoundation, scPRINT) "none outperformed [linear] baselines" for predicting transcriptome changes after perturbations [6]. Zero-shot evaluations show foundation models "perform worse than selecting highly variable genes and using more established methods such as Harmony and scVI in cell type clustering" [7].
- **Multi-omics integration** methods (MOFA, MOFA+) treat different molecular modalities as parallel measurements of the same cell state, despite the fact that "the timescale of (phospho)proteome changes falls into a minute regime, whereas most significant transcriptome changes are observed after hours" [8].

Each puzzle has been addressed within the field as a methodological problem to be solved with better algorithms. The proposed solutions follow a common pattern: incorporate additional data modalities (chromatin accessibility, ATAC-seq, methylation), incorporate prior knowledge (knowledge graphs, motif libraries), or move to more complex architectures (transformers, foundation models, graph neural networks). Some of these solutions help; others do not. The field lacks a structural framework for predicting *which* methodological improvements will help and *why*.

1.2 The OI framework and its C1–C4 conditions

The Observational Incompleteness (OI) framework [1] establishes a characterization theorem for embedded observation. The theorem states: any embedded observer whose dynamics satisfies four conditions — C1 (non-trivial coupling between visible and hidden sectors), C2 (record persistence; the slow bath $\tau_B \gg \tau_S$ one realization), C3 (sufficient hidden memory capacity), and C4 (history readback) — exhibits, when those conditions are realized in the memory-bearing form, accessible non-Markovian dynamics (P-indivisibility being the strictly stronger temporal-indivisibility/backflow diagnostic, not an automatic consequence — `review4_probes.py`): history-dependent transition probabilities arising from information stored in the hidden sector and returned on subsequent interactions. The theorem is scale-independent: it applies to any system with the appropriate architecture, from the memory-bearing dynamics quantum mechanics represents at the cosmological horizon to biological signaling cascades.

The companion Medicine paper [2] establishes that biological memory systems satisfy C1–C4 at multiple molecular layers. The chromatin layer is the most directly relevant for computational biology: DNA methylation patterns, histone modifications, and chromatin-state assignments form a slow hidden sector with substantial τ_B relative to transcriptional dynamics, and the chromatin state space is exponentially large (over $4^{N_{\mathrm{CpG}}}$ for $N_{\mathrm{CpG}} \sim 28$ million CpG sites in the human genome). Cellular dynamics — the trajectory a cell takes through state space during differentiation, perturbation response, or homeostasis — depends on both visible-sector dynamics (transcription, signaling) and hidden-sector state (chromatin memory).

The framework's bioinformatic prediction follows from the theorem: methods that infer cellular dynamics from visible-sector data alone (transcriptome) must systematically miss the contribution of hidden-sector memory. The errors are not random noise; they have a specific structure: directional biases, reduced accuracy in transitions where chromatin state changes, and information-theoretic ceilings on what visible-sector data can recover. Methods that include hidden-sector information explicitly (multi-modal data, knowledge graphs, prior biological information) should systematically outperform pure visible-sector methods on these tasks.

1.3 Thesis and structure

This paper develops the framework's prediction across five active methodological domains in single-cell bioinformatics. Each section follows a common structure: (i) the documented field-level puzzle or failure mode; (ii) the framework's structural explanation in terms of C1–C4 dynamics; (iii) the empirically observed methodological improvements that align with the framework's prediction.

§2 addresses trajectory inference, from RNA velocity's documented failures to multi-omic velocity's improvements. §3 addresses gene regulatory network inference, including the field's "persistent puzzle" of causal methods failing to outperform correlation. §4 addresses perturbation prediction, where foundation models systematically underperform simple baselines. §5 addresses multi-omics integration, where static methods struggle with the timescale hierarchy across modalities. §6 addresses cancer methylome classification, where the framework's substratum-preservation prediction matches the entire field's diagnostic paradigm. §7 develops a new testable prediction about variant pathogenicity stratification for reader/writer/eraser genes. §8 extends the framework's reader-writer disorder class [Medicine §10.7] into bioinformatic patterns. §9 synthesizes the cross-domain findings. §10 catalogs open questions and limits.

The paper's contribution is not new algorithms. It is a structural framework for understanding why current methods fail in specific patterns, and for predicting which methodological improvements will work. The information-theoretic ceiling identified here has practical consequences: it explains why scaling foundation models has not produced the expected improvements, and predicts that future progress requires architectural changes that incorporate hidden-sector information rather than scaling visible-sector models further.

2. Trajectory Inference

2.1 The RNA velocity paradigm and its assumptions

RNA velocity, introduced by La Manno et al. (2018) [9] and operationalized in the velocyto and scVelo tools, infers the direction of cellular state change from the ratio of unspliced to spliced mRNA in single-cell RNA sequencing data. The method's appeal is that it extracts dynamic information from static snapshots: by measuring the relative abundance of pre-mRNA (unspliced, recently transcribed) and mature mRNA (spliced, accumulated), it estimates whether a gene is in an induction phase (more unspliced relative to expected steady state) or repression phase (more spliced relative to expected steady state). Aggregating across genes produces a velocity vector for each cell, which can be visualized as a flow field over the dimensionally-reduced state space.

The original velocyto method [9] assumes (i) constant rates of transcription and degradation, (ii) a global splicing rate shared across genes, (iii) the dynamics has reached steady-state equilibrium in the induction phase, and (iv) gene-wise independence. The scVelo extension [10] relaxes (i) and (iii) but retains the assumption that within-gene dynamics is governed by a closed kinetic equation depending only on current mRNA levels. Both methods treat the gene-state dynamics as Markovian: the probability of state change at time $t + \Delta t$ depends only on the state at time t, not on the cell's history.

2.2 Documented failure modes

Gorin, Pachter, and Yosef (2022) [3] published a detailed analysis of RNA velocity methods in PLOS Computational Biology, demonstrating that scVelo "fails to identify, and even reverses the trajectory" in multiple validated developmental datasets. The reversed trajectories produce qualitatively wrong biological conclusions: cell types identified as progenitors are actually terminal states, and the inferred differentiation direction is opposite to the ground truth. The failures are documented across at least six independent datasets [3, 11–14].

Barile et al. (2021) [11] and subsequent analyses identify the proximate cause: when cell populations have heterogeneous kinetics (different genes operating at different rates) or when the experiment captures cells not in steady state, scVelo's parameter estimates are biased in a way that systematically reverses the inferred direction. The DeepVelo paper [12] notes that "these assumptions are often violated in complex biological systems and bring about limitations in downstream applications, particularly when cell states are partially observed or undergo transcription dynamics more complex than the steady-state pattern."

The pattern is consistent: RNA velocity methods work for simple linear trajectories with homogeneous kinetics and fail for complex trajectories with heterogeneous dynamics. The methodological response has been to develop progressively more flexible parameter estimation (LatentVelo [15], veloVI [16], DeepVelo [12]). These improvements help marginally but do not eliminate the failure mode: the underlying issue is structural, not parametric.

2.3 The framework's structural explanation

The OI framework predicts the RNA velocity failure mode structurally. Gene expression in single cells is governed by the joint dynamics of the transcriptional machinery (visible sector) and the chromatin state (hidden sector). Chromatin state determines which genes are accessible for transcription, at what rates, and with what kinetic parameters; chromatin state itself evolves on slower timescales than transcription, with histone modifications turning over on minutes-to-hours scales and DNA methylation on days-to-weeks scales [Medicine §9.1]. The full dynamics is non-Markovian in the gene-expression representation: knowing the current spliced/unspliced ratios is insufficient to predict the next state, because chromatin memory contributes information not captured by mRNA levels alone.

The C1–C4 architecture applies directly: gene transcription (fast, visible) is coupled (C1) to chromatin state (persistent — slow, C2) with exponentially large chromatin state space (C3), and the chromatin marks transcription lays down modulate subsequent transcription — the coupling reads back what it wrote (C4). With all four realized, the framework's characterization theorem [1] then implies that RNA velocity inferred from gene-expression data alone is memory-bearing — accessibly non-Markovian, with P-indivisibility the stronger diagnostic requiring a measured witness (`review4_probes.py`): the marginal dynamics on the gene-expression subspace cannot be described by a Markovian kinetic equation, and methods assuming such description will produce biased estimates.

The bias is not arbitrary. Its sign depends on the relationship between chromatin state and transcriptional kinetics. When chromatin transitions are slow relative to the measurement timescale (typical for differentiation), the chromatin state is approximately frozen, and gene-expression dynamics within a chromatin state can be approximately Markovian — RNA velocity works. When chromatin transitions are comparable to or faster than the measurement timescale (typical for cell-fate transitions, perturbation responses, and complex differentiation), the chromatin state changes substantially within the observation window, and gene-expression dynamics is non-Markovian — RNA velocity fails, with specific systematic biases depending on the chromatin transition pattern.

This matches the empirical pattern in Gorin et al. and the surrounding literature: simple linear trajectories with homogeneous kinetics succeed; complex trajectories with heterogeneous dynamics fail. The framework explains why this is the structural pattern rather than a methodological accident.

2.4 Multi-omic velocity as empirical convergence

The methodological response that does work is to include chromatin accessibility data explicitly. MultiVelo [17], introduced by Li et al. (2022) in Nature Biotechnology, "extends the RNA velocity framework to incorporate epigenomic data" and "improves the accuracy of cell fate prediction compared to velocity estimates from RNA only." The model includes both transcriptional dynamics (spliced/unspliced ratios) and chromatin accessibility dynamics, fitting a coupled differential equation system. By estimating switch times for chromatin opening/closing alongside transcription rates, MultiVelo recovers two distinct classes of genes (chromatin closes before vs. after transcription ceases) and four types of cell states distinguishing coupled and decoupled chromatin-transcription dynamics.

The methodological trajectory has continued. mmVelo [18] (2024) extends multi-modal velocity estimation to single-peak resolution rather than gene-aggregated chromatin signals. MoFlow [19] (Nature Communications 2025) integrates multi-omic data for "locally adaptive, single-cell-resolved velocities that illuminate fine-scale regulatory dynamics." Each successive method incorporates more hidden-sector information (chromatin accessibility at higher resolution, additional modalities, locally adaptive dynamics) and improves performance.

This trajectory matches the framework's prediction. The methods that work are the methods that include the hidden sector explicitly; the methods that fail are the methods that assume Markovian dynamics on the visible sector alone. The empirical convergence over the 2022–2025 period — from RNA-only methods through gene-level multi-omic methods to single-peak resolution multi-omic methods — represents the field's gradual adoption of the framework's structural prediction without explicit reference to the underlying theorem.

2.5 Practical consequences

The framework provides three practical predictions for trajectory inference methodology:

(P1) RNA-only trajectory methods will continue to fail for systems with substantial chromatin dynamics during the observation window. Parameter refinements within the RNA-only paradigm (improved kinetic models, deeper neural networks on transcriptome alone) cannot fix this; the limit is information-theoretic.

(P2) Multi-modal methods that include chromatin accessibility will outperform RNA-only methods on differentiation, perturbation response, and cell-fate transition tasks. The improvement should scale with the relative timescale of chromatin transitions to transcription: faster chromatin dynamics relative to transcription means larger improvement from including chromatin data.

(P3) The optimal data modality combination depends on the timescale of the biological process being studied. For processes with τ_B at the histone-modification timescale (minutes-to-hours), histone-mark profiling (CUT&Tag, CUT&RUN) is more informative than DNA methylation. For processes at the methylation timescale (days-to-weeks), methylation profiling is more informative. For long-term memory (developmental trajectories spanning generations), DNA methylation and chromatin architecture (Hi-C) become the dominant relevant modalities. Mixing modalities should be calibrated to the biological process's characteristic τ_B rather than included indiscriminately.

The third prediction is testable through benchmarking: a study comparing multi-modal velocity methods across biological processes with different characteristic timescales should show the relative performance of each modality combination tracking the framework's prediction.

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3. Gene Regulatory Network Inference

3.1 The GRN inference paradigm and the "persistent puzzle"

Gene regulatory networks (GRNs) describe the directed influence of transcription factors on target genes. Inferring GRNs from single-cell expression data is a foundational task in computational biology, with applications ranging from drug-target identification to mechanistic understanding of differentiation programs. The methodology has produced a series of method generations: correlation-based methods (Pearson, mutual information), ensemble methods (GENIE3, GRNBoost2), expression-cell-type specific methods (SCENIC), causal methods (PC algorithm, NOTEARS, DeepSEM), and most recently deep-learning foundation models (scGPT, Geneformer) repurposed for GRN inference.

Despite the methodological investment, the empirical state of GRN inference accuracy is sobering. Yuan et al. (Nature Biotechnology 2024) [20] state: "Despite extensive efforts, GRN inference accuracy has remained disappointingly low, marginally exceeding random predictions." Passemiers et al. (2025) [5] document a "persistent puzzle which casts doubt on the value of causality for this task": causal methods consistently fail to outperform correlation-based baselines in realistic benchmarks, despite their theoretical advantages.

Multiple comprehensive benchmarking studies have confirmed this pattern. McCalla et al. [21] evaluated 13 GRN inference methods on 7 published datasets using diverse gold standards; the methods rank similarly across gold standards, with no method showing decisive advantage. Nguyen et al. [22] benchmarked 15 methods on 139 simulated datasets and found extensive failure modes related to dropout rate and sparsity. The conclusion across the literature is consistent: GRN inference from single-cell data is hard, and current methods perform near baseline.

3.2 Why correlation outperforms causation: the framework's prediction

The persistent puzzle has a natural framework explanation. Causal methods assume that the joint distribution over gene expression captures the system's dynamics: from sufficient samples, the conditional independence structure reveals the causal graph. This assumption is valid for systems where the visible-sector dynamics is autonomous — where current gene expression states fully determine future states up to noise.

The framework predicts this assumption fails for cellular systems. Gene regulation depends on chromatin state, transcription factor protein levels, post-translational modifications, and other variables not captured in scRNA-seq data. The marginal distribution over gene expression is the result of integrating out these hidden variables, and the conditional independence structure of the marginal distribution does *not* match the causal graph of the underlying dynamics. Causal inference methods recover the marginal structure, not the underlying causal structure, because the data does not contain enough information to distinguish them.

Correlation-based methods make a weaker claim: they identify pairs of genes whose expression covaries, without claiming causal direction. The covariance pattern is robust to hidden-variable integration in a way that the causal structure is not. Correlation-based methods perform competitively because they target a tractable quantity (the marginal covariance) rather than an intractable one (the marginal causal structure).

This is the framework's prediction: in non-Markovian systems where the visible sector does not contain the full dynamical information, correlation-based methods will systematically outperform causal methods that assume Markovian dynamics on the visible sector. The "persistent puzzle" is the empirical signature of this structural prediction.

3.3 Multi-modal methods as the predicted improvement

The methodological response that improves GRN inference accuracy is to incorporate additional data modalities. SCENIC+ [23], introduced by González-Blas et al. (Nature Methods 2023), extends the original SCENIC pipeline by incorporating chromatin accessibility data (scATAC-seq) alongside gene expression. The method identifies enhancer-driven gene regulatory networks by predicting genomic enhancers, linking them to candidate transcription factors via motif analysis, and linking enhancers to target genes via expression correlation across cells. SCENIC+ improves substantially over RNA-only SCENIC and over earlier methods that used only expression data.

LINGER [24], introduced by Yuan et al. (Nature Biotechnology 2024), takes the multi-modal approach further by incorporating "atlas-scale external bulk data across diverse cellular contexts and prior knowledge of transcription factor motifs as a manifold regularization." The method combines paired scRNA-seq + scATAC-seq data with external bulk reference data, encoding prior biological knowledge about transcription factor binding preferences. The performance improvement is substantial: LINGER's GRN predictions outperform SCENIC+ and earlier methods across multiple benchmarks.

The trajectory matches the framework's prediction. RNA-only methods are near random baseline. Adding chromatin accessibility (substratum-level visible) helps. Adding prior biological knowledge (encoded hidden structure) helps further. Each addition incorporates more hidden-sector information, and each addition improves accuracy. The framework's structural prediction — that GRN inference accuracy scales with how much hidden-sector information is included in the data — is empirically realized in the methodological development trajectory of the past five years.

3.4 Information-theoretic ceiling

The framework predicts more than just a relative improvement from multimodal data. It predicts an information-theoretic ceiling: no method using only scRNA-seq data can recover the true causal GRN, regardless of model complexity or training data scale. This is the structural content of P-indivisibility: information that has been integrated out cannot be recovered from the marginal observations.

The recent attempts to apply deep-learning foundation models to GRN inference test this prediction directly. scGPT [25] and Geneformer [26], trained on tens of millions of single cells, attempt to learn gene-gene relationships from transcriptome data alone. Multiple independent evaluations have shown that these models do not improve substantially over simpler correlation-based methods on GRN inference (§4 develops this for perturbation prediction; similar results hold for GRN tasks). The foundation models' attention patterns "capture co-expression rather than unique regulatory signal" [27] — they recover the marginal correlation structure that simpler methods also recover, without accessing the underlying regulatory causal structure.

This is the information-theoretic ceiling in action. Scaling transcriptome-only foundation models does not break through the ceiling; the ceiling is structural, not computational. The methodological roadmap forward requires architectural changes that incorporate hidden-sector information, not further scaling of visible-sector models.

3.5 Practical consequences

Three framework-grounded predictions for GRN inference methodology:

(P4) GRN inference from scRNA-seq alone is near its information-theoretic ceiling. Further methodological development that stays within RNA-only data will produce marginal improvements, not the order-of-magnitude gains the field has been seeking.

(P5) Multi-modal methods (RNA + ATAC, RNA + methylation, RNA + Hi-C) will systematically outperform RNA-only methods. The relative gain depends on which hidden-sector modality is most informative for the specific regulatory structure being inferred.

(P6) Foundation models applied to scRNA-seq alone will not exceed the information-theoretic ceiling, regardless of scale. Foundation models trained on multi-modal data (paired RNA + ATAC, multi-omics atlases) should outperform single-modality foundation models on GRN inference tasks. This is testable: compare scGPT trained on RNA-only versus a foundation model of comparable scale trained on multi-modal data.

The third prediction is testable in the near term as multi-modal foundation models become available.

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4. Perturbation Prediction

4.1 The perturbation prediction paradigm

Single-cell perturbation experiments — CRISPR knockouts (Perturb-seq, CROP-seq), CRISPRa/i activation/repression (Replogle et al. 2022 [28]), drug treatments (sci-Plex), and combinatorial perturbations (Norman et al. 2019 [29]) — produce datasets where individual cells are labeled by the perturbation they received and the resulting transcriptional state is measured. The computational task is to predict the transcriptional state for *unseen* perturbations: combinations of perturbations not in the training data, perturbations applied to new cell types, or perturbations with new drugs.

This task has attracted substantial methodological investment. Generative models (CPA [30], chemCPA [31]) learn compositional representations of perturbation effects. Graph neural networks (GEARS [32]) incorporate Gene Ontology and protein-protein interaction graphs as priors. Optimal transport methods (CellOT [33]) match unperturbed and perturbed distributions. Most recently, single-cell foundation models (scGPT [25], Geneformer [26], scFoundation [34], scPRINT) have been adapted for perturbation prediction tasks, with the expectation that pretraining on tens of millions of cells would enable strong zero-shot and few-shot performance.

4.2 The empirical failure of deep learning on perturbation prediction

Ahlmann-Eltze et al. (Nature Methods 2025) [6] published a direct evaluation of this expectation. The study compared five foundation models — scGPT, Geneformer, scFoundation, scPRINT, and Universal Cell Embeddings — and two additional deep-learning models against deliberately simple linear baselines for predicting transcriptome changes after single or double perturbations. The result: “None outperformed the baselines.” The simplest baseline — predicting the mean response across the training set — was competitive with or superior to every deep-learning method evaluated.

The result has been replicated. The geneRNIB benchmark (Passemiers et al. 2025) [5] confirms that deep-learning methods do not consistently outperform linear baselines on perturbation prediction tasks. Independent analyses of scGPT’s attention mechanisms [27] show that the attention patterns “capture co-expression rather than unique regulatory signal” — the model has learned the gene-gene correlation structure that simpler methods also capture, without accessing additional information.

The pattern is structural, not technical. The foundation models are well-implemented, well-trained, and demonstrably effective on tasks where the visible-sector data contains sufficient information (cell type annotation with appropriate fine-tuning, batch correction in some settings). The failure is specific to tasks where the missing information matters: predicting how cells will respond to perturbations not seen in training requires knowing what the cell can become, which depends on chromatin state, transcription factor activity, and other variables not in the transcriptome.

4.3 The framework’s structural prediction

The OI framework’s information-theoretic constraint produces a hard upper bound on what transcriptome-only perturbation models can achieve. The argument is direct: the cell’s response to a perturbation depends on (i) the current transcriptional state, (ii) the chromatin state determining which genes are accessible for activation, (iii) the protein levels of transcription factors not visible in mRNA data, and (iv) the metabolic state. Of these, (i) is in the data and (ii)–(iv) are not. The function mapping perturbation $\times$ cell state to response state is therefore a function of more variables than the data contains; the mapping cannot be learned from data alone, no matter how much data is available, because the inverse problem is structurally under-determined.

The constraint is information-theoretic in the strict sense: the mutual information between the visible-sector data and the perturbation response is bounded above by a quantity that does not scale with dataset size. Scaling the model and the data improves estimation of the conditional mean response (which is why deep-learning models do achieve some performance), but cannot improve prediction of the response’s *variance*, *outlier distribution*, or *cell-specific deviations* — all of which depend on hidden-sector variables.

The Ahlmann-Eltze result has a specific structural interpretation in this framework: when the baseline (predict the mean) is competitive with foundation models, it indicates that the foundation models are also predicting close to the conditional mean response, because the conditional mean is the limit of what transcriptome data supports. Deep learning architectures do not extract additional information from the visible sector because there is no additional information to extract.

4.4 Knowledge-graph methods as the predicted improvement

The methods that do exceed simple baselines incorporate explicit hidden-sector information through external knowledge. GEARS [32] uses Gene Ontology graphs and protein-protein interaction networks as structural priors during training; the prior knowledge constrains the prediction in ways that pure data-driven learning cannot. PerturbNet [35] (Mol Syst Biol 2025) achieves the highest R^2 values across multiple benchmarks by separating the drug representation network (trained on chemical structure) from the cell representation network (trained on expression), allowing the model to compose information from disjoint training sources. Both methods incorporate information that is not in the scRNA-seq data — chemical structure, ontological gene relationships, prior regulatory knowledge — and both outperform pure data-driven foundation models.

The methodological trajectory is consistent with the framework: the methods that work are the methods that include hidden-sector information explicitly through external knowledge sources, and the methods that fail are the methods that try to extract that information from transcriptome data alone. The "deep learning fails to beat baselines" puzzle is the framework's information-theoretic ceiling made empirically manifest.

4.5 Compositional generalization and what scaling cannot fix

A subtler prediction from the framework concerns compositional generalization — the ability to predict combinations of perturbations from training data containing only single perturbations. Standard intuition is that with enough data and a sufficiently flexible model, compositional structure should be learnable. The framework's prediction is more restrictive: compositional generalization works only when the perturbations act on visible-sector variables independently, and fails when perturbations interact through hidden-sector variables (chromatin state, transcription factor competition, etc.).

This is testable. Norman et al. (2019) [29] published a combinatorial perturbation dataset with 287 single-gene knockdowns and 131 pairs of knockdowns. The framework predicts that pairs where both genes act on the same chromatin domain (likely to interact through hidden-sector variables) should be harder to predict from single-gene training data than pairs where the genes act on disjoint pathways. The empirical pattern in compositional generalization benchmarks should track the chromatin co-localization of the perturbed genes, not just their expression patterns. This prediction has not been directly tested in the existing benchmarking literature.

4.6 Practical consequences

Four framework-grounded predictions for perturbation prediction methodology:

(P7) Transcriptome-only foundation models will not exceed the conditional-mean baseline for perturbation prediction, regardless of model scale or training data size. The ceiling is information-theoretic and not addressable through more pretraining.

(P8) Methods incorporating chromatin accessibility data (paired Perturb-seq + ATAC-seq) will systematically outperform RNA-only methods on perturbation prediction. The improvement should be largest for perturbations affecting chromatin-state transitions (knockdowns of chromatin modifiers, transcription factors with autoregulatory loops).

(P9) Methods incorporating prior knowledge (knowledge graphs, motif libraries, regulatory network priors) will systematically outperform pure data-driven methods. The improvement scales with how well the prior captures the actual hidden-sector structure relevant to the prediction task.

(P10) Compositional generalization (predicting double perturbations from single-perturbation training) will be systematically harder for perturbations co-localized at the chromatin level than for perturbations acting on disjoint pathways. This is testable through benchmarking on Norman et al.'s combinatorial dataset stratified by chromatin co-localization of the perturbed gene pairs.

The fourth prediction is particularly diagnostic. If it holds, it provides direct empirical confirmation that the framework's structural picture (chromatin as hidden sector mediating perturbation interactions) is the right explanation for compositional generalization failure modes.

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5. Multi-Omics Integration

5.1 The multi-omics integration paradigm

Single-cell multi-omics technologies measure multiple molecular modalities in the same cell: G&T-seq (genome + transcriptome), scNMT-seq (chromatin accessibility + transcriptome + methylation), CITE-seq (transcriptome + surface protein), QuRIE-seq (transcriptome + intracellular phosphoprotein), and increasingly multimodal Perturb-seq variants. The data architecture is: N cells $\times$ M modalities $\times$ G_m features per modality, with the modalities measuring different aspects of cellular state on different timescales.

The standard computational task is integration: combining information across modalities to produce unified cell embeddings, joint trajectories, and integrated regulatory networks. The principal methods are Multi-Omics Factor Analysis (MOFA [36] and MOFA+ [37]), Seurat's weighted nearest neighbor (WNN) [38], LIGER [39], and totalVI [40]. These methods share a common assumption: different modalities are parallel measurements of the same underlying cell state, and integration combines them into a single low-dimensional latent space.

5.2 The timescale problem

The parallel-measurements assumption fails when modalities measure different memory layers operating on different timescales. The QuRIE-seq paper [8] explicitly states: "The timescale of (phospho)proteome changes falls into a minute regime, whereas most significant transcriptome changes are observed after hours." The same cells, measured at the same instant, contain information from different points in their dynamical history — phosphoprotein measurements reflect minute-scale signaling state, transcriptome measurements reflect hour-scale gene expression state, and methylation measurements reflect day-to-week-scale chromatin state.

The framework's perspective on this is direct: each modality measures a different memory layer with a different τ_B. The visible-sector / hidden-sector partition is modality-dependent. From the perspective of phosphoprotein dynamics, the transcriptome state is part of the slow hidden sector (chromatin sets transcriptional rates, which set protein production rates). From the perspective of methylation dynamics, the transcriptome is part of the fast visible sector (transcription happens fast on the methylation timescale). The static factor-analysis assumption — that all modalities reflect the same cell state — misses this hierarchical structure.

The empirical consequences are documentable. Static integration methods produce embeddings that are dominated by the modality with the largest dynamic range during the measurement window — typically transcriptome — and the other modalities contribute mostly as confirmation of the transcriptome-derived structure rather than as independent information. The integration improves accuracy modestly compared to single-modality methods, but does not fully exploit the temporal information present in the multi-modal data.

5.3 Emerging timescale-aware methods

Recent methodological work has begun to address the timescale problem explicitly. The QuRIE-seq paper [8] used time-series sampling combined with MOFA+ factors to capture dynamics across modalities. Heyne et al. (2024) [41] introduced pseudotemporal integration of single-cell RNA-Seq and single-cell proteomics by mass spectrometry, building cell-state trajectories separately for each modality and aligning them through shared markers. The approach yields "time-dependent, nonlinear relationships between transcript and protein levels" that static methods cannot recover.

The chromatin velocity literature (§2) is a special case of timescale-aware multi-omics integration: MultiVelo [17] explicitly models chromatin and transcription as coupled dynamical systems with different switch times. The success of MultiVelo over scVelo is the timescale-aware version of the framework's prediction — using temporal hierarchy in the integration produces better dynamical estimates.

The framework predicts further methodological development along these lines will succeed. Specifically, integration methods that include explicit lag structure between modalities (transcriptome at time t reflecting chromatin state at time $t - \tau_B$) should outperform methods that treat modalities as instantaneous parallel measurements. The lag structure should be biologically calibrated to the characteristic τ_B of each modality.

5.4 The hierarchy of biological memory layers

The framework provides a specific prediction for the timescale hierarchy across biological modalities, ordered from fastest to slowest:

Modality	Typical τ_B	Memory layer
Phosphoproteins	seconds–minutes	Signaling state
Metabolites	seconds–minutes	Metabolic state
Transcriptome	minutes–hours	Gene expression state
Histone acetylation	minutes–hours	Active chromatin state
Histone methylation	hours–days	Stable chromatin state
Proteome	hours–days	Translation state
Chromatin accessibility	hours–days	Regulatory element state
DNA methylation	days–weeks	Stable epigenetic state
Genome (sequence)	$\to \infty$	Heritable state
Chromatin 3D architecture (Hi-C)	hours–generations	Genome organization

These timescales are approximate and tissue-dependent, but the ordering is robust. The framework predicts that integration methods that respect this ordering — treating slower modalities as constraints on faster modalities — will systematically outperform methods that treat them symmetrically.

5.5 Static factor analysis under the framework

MOFA and MOFA+ are widely used and produce useful cell embeddings. They do not, however, capture the temporal hierarchy across modalities. The framework's prediction is that MOFA-style methods extract the *static covariance structure* shared across modalities — what is common to all measurements of a cell at one instant — but miss the *dynamical relationships* between modalities — how chromatin state at one time predicts transcriptome state at a later time. The static covariance is informative but incomplete; the dynamical relationships contain additional information that static methods cannot access.

This is testable. A direct comparison of MOFA+ integration with a timescale-aware method on time-series data should show the timescale-aware method recovering dynamical relationships that MOFA misses. The QuRIE-seq paper [8] and subsequent time-series multi-omics studies provide the data substrate for such comparisons; the comparison has not been systematically performed across the methodological literature.

5.6 Practical consequences

Three framework-grounded predictions for multi-omics integration:

(P11) Static factor-analysis integration methods (MOFA, MOFA+, LIGER, totalVI) systematically underutilize the temporal information present in multimodal data. The performance gap relative to timescale-aware methods should be largest in dynamic biological processes (perturbation responses, differentiation) and smallest in steady-state systems.

(P12) Timescale-aware integration methods will systematically outperform static methods on prediction tasks involving cellular dynamics. The improvement should scale with the ratio between the slowest and fastest modality timescales in the data: large ratios (transcriptome + methylation) should show large improvements; small ratios (transcriptome + chromatin accessibility, since both operate on hour-scale timescales) should show smaller improvements.

(P13) The optimal modality combination for a given biological question depends on the characteristic timescale of the process. Studies of fast signaling dynamics should prioritize phosphoproteome + transcriptome; studies of stable cell-fate decisions should prioritize methylome + transcriptome; studies of developmental trajectories should include chromatin architecture (Hi-C) as the slowest layer.

The framework provides a structural rationale for the multi-omics integration practices that bioinformaticians are converging on through empirical experience. Rather than treating multi-omics as a generic improvement, the framework predicts which modality combinations matter for which biological questions, and which integration methods will succeed.

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6. Cancer Methylome Classification

6.1 The methylation-based tumor classification paradigm

DNA methylation profiling has emerged over the past decade as a central diagnostic tool in neuro-oncology and increasingly across cancer types. The Heidelberg CNS Tumor Classifier, introduced by Capper et al. (Nature 2018) [42] with version v11 trained on 2,801 reference samples, classified central nervous system tumors into 91 methylation classes. Version 12.8 (Cancer Cell 2025) [43] expanded this to 184 subclasses trained on 7,495 methylation profiles, achieving 95% subclass-level accuracy with well-calibrated probabilistic scores. Methylation-based classification is now incorporated into the WHO Classification of CNS Tumors (CNS5, 2021) as a desirable or essential method for accurate diagnosis of multiple tumor types [44].

The classifier reads DNA methylation patterns at hundreds of thousands of CpG sites (Illumina EPIC 850K/930K array) and assigns the sample to a reference tumor class based on similarity to a curated training set. The methylation signature is informative because, as Capper et al. (2018) [42] explicitly note, "the genome-wide DNA methylation pattern (the methylome) likely represents a combination of both the cell of origin and somatically acquired DNA methylation changes." The cell-of-origin component dominates the classification: methylation patterns reliably identify the developmental lineage from which the tumor arose, even when the tumor's morphology, histology, or transcriptional state has diverged substantially from the lineage's normal pattern.

6.2 Methylome arrest in cancer

A critical observation in the methylation-classification literature is that DNA methylation patterns are *relatively stable* in cancer cells once established. Capper et al. (2018) [42]: "the dynamics of the methylome crucial for cellular differentiation in non-transformed cells seem to be somewhat arrested in cancer cells and DNA methylation signatures likely remain relatively stable during the course of disease." The methylome is not dynamic in the same way as the transcriptome or proteome; it is stable enough to function as a reliable diagnostic marker across disease progression, treatment, and recurrence.

This stability is what makes methylation classification work clinically. If cancer methylomes were as dynamic as cancer transcriptomes, a single methylation profile could not be reliably associated with a tumor class — the profile would shift with disease progression and treatment, just as transcriptional profiles do. The methylome's relative stability across disease course is the practical foundation of the entire methylation-classification paradigm.

6.3 The substratum-emergent operator distinction in cancer

The framework's predicted pattern for cancer is the substratum-emergent operator distinction [Medicine §9.4] applied to oncology: the substratum-level memory operator (DNA methylation pattern) is preserved with high fidelity reflecting the cell of origin, while the emergent-level operator (cell behavior, proliferation, differentiation, transcriptional phenotype) is dysregulated. The two operators dissociate in cancer in a specific direction: substratum preserved, emergent dysregulated.

This is the structural pattern the methylation-classification field has empirically established. The 95% subclass-level accuracy of methylation-based classification reflects the high fidelity of the substratum-level memory; the fact that the same methylation patterns associate with diverse histological appearances, transcriptional states, and clinical behaviors reflects the dysregulation of the emergent operator. The framework's prediction matches the field's diagnostic paradigm precisely.

This same pattern operates across scales: sphalerons and baryon-number violation in particle physics, iPSC reprogramming residuals in stem cell biology [Medicine §9.4], and methylation-classifier-based diagnosis in oncology are all instances of the broader substratum-emergent operator distinction observed in foundational physics [SM §8.7] — where operators conserved at the substratum (lattice) level can be violated at the emergent (effective field theory) level. The pattern is unified across scales: the substratum carries information that the emergent operator can violate, and methylation classification in oncology operationalizes this for clinical diagnostics.

6.4 Unclassifiable tumors and substratum disruption

A specific empirical pattern in methylation classification is informative for the framework. Drexler et al. (2024) [45] document that approximately 10–15% of CNS tumor cases submitted to the Heidelberg classifier are "unclassifiable" — they do not match any reference methylation class with confidence above the diagnostic threshold (calibrated score ≥ 0.9). These unclassifiable cases have systematically distinct features: lower tumor purity, younger patient age, recurrent tumor tissue post-radiotherapy, and significantly worse clinical outcomes including longer time to treatment initiation and less favorable survival in IDH wildtype glioblastomas.

The framework's prediction for unclassifiable tumors is that they are cases where the substratum-level memory operator has been disrupted, not just the emergent operator. The disrupted substratum produces a methylation pattern that no longer reflects a single cell of origin, and the classifier cannot map it to a reference class. The empirical features of unclassifiable cases match this prediction: lower tumor purity (contributions from multiple cell types altering the methylation pattern), post-radiotherapy recurrence (DNA damage altering methylation maintenance), and younger age (less time for methylation to stabilize into a clear pattern).

This is testable beyond clinical features. The framework predicts that unclassifiable tumors should show specifically altered methylation patterns at maintenance sites (DNMT1 binding sites, hemi-methylated regions) and at chromatin boundary elements, reflecting disruption of the methylation maintenance machinery rather than tumor-type-specific methylation differences. The hypothesis is testable in the existing Heidelberg dataset by stratifying classifiable versus unclassifiable cases on machinery-site versus content-site methylation patterns.

6.5 Methylation classification beyond CNS tumors

The methylation-classification paradigm has been extended to other cancer types: sarcoma classifiers [46], leukemia classifiers, and pan-cancer methylation classifiers [47]. The performance pattern is consistent: high accuracy for tumor type identification, with the methylation signature carrying cell-of-origin information that survives the tumor's emergent dysregulation.

The framework predicts that this paradigm will continue to expand across cancer types, and that the boundary cases — cancer types where methylation classification has lower accuracy — will be those where the substratum-level memory is more readily disrupted in the cancer state. Highly mutated cancers with extensive copy number aberrations (which can disrupt chromatin maintenance machinery) should be less amenable to methylation classification than cancers with stable genomes. This is a specific testable prediction about cross-cancer methylation classifier performance.

6.6 Practical consequences

Three framework-grounded predictions for cancer methylome bioinformatics:

(P14) The methylation-classification paradigm will continue to extend to cancer types where the substratum-level memory is preserved during tumorigenesis. Cancers with stable genomes (most pediatric CNS tumors, well-differentiated sarcomas) will achieve high classifier accuracy; cancers with extensive chromatin maintenance machinery mutations will achieve lower accuracy.

(P15) Unclassifiable tumors are not a single category but split into two distinct groups: (i) cancers from rare cell-of-origin lineages not represented in the training data (the standard interpretation), and (ii) cancers where the substratum-level memory has been disrupted (the framework's prediction). The two groups should have distinguishable methylation patterns at maintenance versus content sites.

(P16) Methylation-based classifiers can be improved by including chromatin maintenance machinery integrity (mutations or copy number alterations in DNMT1, DNMT3A/B, TET1/2/3, etc.) as a covariate. The framework predicts that tumors with intact maintenance machinery should be more reliably classifiable than tumors with disrupted machinery, at given tumor purity.

The third prediction is testable through reanalysis of the existing Heidelberg cohort with maintenance machinery integrity as a stratification variable.

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7. Variant Pathogenicity for Reader/Writer/Eraser Genes

7.1 Variant effect prediction methods

Predicting the functional impact of missense variants on protein-coding genes is a core task in clinical genetics. Methods range from sequence-conservation methods (SIFT [48], PolyPhen-2 [49]) through structure-aware methods (CADD [50], REVEL [51]) to AI-based methods (AlphaMissense [52]). AlphaMissense, introduced by Cheng et al. (Science 2023) [52], applies the AlphaFold2 architecture [53] to variant effect prediction and achieves state-of-the-art performance across multiple benchmarks, providing pathogenicity scores for every possible single-amino-acid substitution in the human proteome [54].

The methods operate on a common assumption: variant pathogenicity is determined by the variant's impact on the protein it directly affects. Conservation features capture evolutionary intolerance to variation at the residue; structural features capture the variant's disruption of folding or function; AI features capture the integration of structural and conservation information at a higher level. The methods do not, in general, account for the variant's *functional role* in the broader regulatory architecture — whether the affected gene is a substrate gene, a reader of epigenetic marks, a writer that establishes marks, or an eraser that removes marks.

7.2 The framework's prediction

The Medicine companion paper [Medicine §10.7] establishes the reader-writer disorder class: genetic disorders caused by loss-of-function mutations in readers, writers, or erasers of epigenetic memory (Rett syndrome via MECP2, Fragile X via FMR1, Angelman via UBE3A, Kabuki via KMT2D, ATR-X via ATRX). These disorders have substantially different phenotypic features than disorders affecting substrate genes directly: wider therapeutic windows, demonstrated reversibility in animal models (Guy et al. 2007 [55] for Rett), and graded severity correlated with residual machinery function.

The framework predicts that variant pathogenicity prediction should be systematically different for variants in reader/writer/eraser machinery versus variants in substrate genes. Specifically:

(i) Variants in machinery genes should show *graded* pathogenicity across mutations within the same gene, reflecting the gene's dose-dependent function in maintaining substratum-level memory across the genome. Different MECP2 mutations produce different Rett severity precisely because they confer different degrees of residual reading function [56].

(ii) The relationship between variant pathogenicity score and clinical severity should differ in slope between machinery genes and substrate genes. For substrate genes, loss-of-function correlates directly with phenotype severity (a missing enzyme produces a deficient pathway). For machinery genes, loss-of-function correlates with the *number of downstream targets affected*, which can be partially compensated by residual machinery function — producing a flatter pathogenicity-severity slope.

(iii) The relative performance of variant effect predictors on machinery genes versus substrate genes should differ. Methods trained predominantly on substrate genes (where the conservation-pathogenicity relationship is well-calibrated) should systematically underperform on machinery genes (where the relationship is mediated by downstream effects on chromatin state).

7.3 The proposed empirical test

The framework's predictions can be tested through reanalysis of existing AlphaMissense predictions stratified by gene functional role. The protocol:

1. Classify genes into three categories: substrate genes (encoding enzymes, transporters, structural proteins, ligands, receptors), machinery genes (encoding epigenetic readers, writers, erasers, chromatin remodelers, transcription factors), and architectural genes (encoding scaffolds and connecting structures). Standard ontologies (GO Molecular Function, ChromBase, Eukaryotic Linear Motif database) provide the classification.
2. For each gene category, examine the distribution of AlphaMissense pathogenicity scores conditional on ClinVar pathogenicity assignment.
3. Test prediction (iii): machinery gene scores should show poorer correlation with ClinVar assignments than substrate gene scores, at matched evolutionary conservation level.
4. Test prediction (i): within machinery genes, the variance of pathogenicity scores for clinically observed variants should be higher than within substrate genes, reflecting the graded relationship between machinery function and phenotype.
5. Test prediction (ii): correlate variant pathogenicity score with phenotype severity (where available from clinical literature) and compare slopes across gene categories.

The test is computationally tractable using existing public data (AlphaMissense predictions on Zenodo, ClinVar, gene ontology databases). Negative results would refute the framework's prediction; positive results would establish a new structural pattern in variant effect prediction with practical clinical implications.

7.4 Implications for clinical variant interpretation

If the framework's predictions are confirmed, variant interpretation in reader-writer disorders should incorporate gene functional role as a covariate. Current clinical variant interpretation (ACMG/AMP guidelines [57]) treats all genes symmetrically: variants are scored on evolutionary conservation, computational predictions, segregation data, functional studies, and population frequency, without distinguishing the gene's functional role in the regulatory architecture. The framework predicts this symmetric treatment systematically misclassifies variants in machinery genes — both overcalling and undercalling depending on the specific gene's network position.

A revised approach would (i) flag machinery-gene variants for additional scrutiny, recognizing that pathogenicity for these genes is mediated by downstream effects rather than direct functional loss; (ii) recommend functional assays measuring residual machinery activity rather than relying on prediction scores alone; and (iii) interpret machinery-gene variants in the context of the patient's epigenetic state, since the same variant may produce different phenotypes depending on what substrate it acts on. These recommendations align with emerging clinical practice for Rett-class disorders [58] but have not been generalized to the full machinery-gene category.

7.5 Practical consequences

Three framework-grounded predictions for variant effect prediction methodology:

(P17) AlphaMissense and similar methods systematically miscalibrate pathogenicity scores for reader/writer/eraser genes relative to substrate genes. The miscalibration is testable through stratified reanalysis of existing predictions.

(P18) Variant effect prediction methods trained specifically on machinery-gene variant data should outperform general-purpose methods on this gene category. The methodological recommendation is to develop machinery-gene-specific predictors that incorporate the gene's network position and downstream regulatory targets.

(P19) Clinical variant interpretation should incorporate gene functional role (substrate vs. machinery vs. architectural) as a structural feature affecting how computational pathogenicity scores are weighted. The framework predicts machinery-gene variants require additional functional assay evidence beyond computational scores.

The third prediction has direct clinical implications and is testable through retrospective audit of diagnostic genetic testing outcomes stratified by gene category.

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8. Reader-Writer Disorder Bioinformatics

8.1 The framework's reader-writer disorder class

The Medicine paper [Medicine §10.7] introduces the reader-writer disorder class: genetic disorders caused by loss-of-function mutations in epigenetic readers (MECP2 in Rett syndrome, FMR1 in Fragile X), writers (UBE3A in Angelman syndrome, KMT2D in Kabuki syndrome), or remodelers (ATRX in ATR-X syndrome). These disorders share a structural feature: the substratum-level memory operator (DNA methylation, histone modifications) is largely preserved in patients, while the trace-out machinery that translates substrate state into cellular function is disrupted. The framework predicts that this structural feature produces specific bioinformatic patterns that distinguish reader-writer disorders from substrate-affecting genetic disorders.

8.2 Bioinformatic patterns in Rett syndrome

Rett syndrome provides the most directly testable case. MECP2 is a methyl-CpG-binding protein — a reader of DNA methylation. Patients carry loss-of-function MECP2 mutations; the DNA methylation substrate they read is generally intact [59, 60]. The framework predicts: in Rett patient samples, DNA methylation patterns should be approximately normal (substratum preserved), while gene expression and chromatin state should be altered (emergent operator disrupted).

The empirical literature is consistent with this prediction. Picard et al. (2021) [60] review the transcriptomic and epigenomic landscape in Rett: while large-scale alterations of the epigenome have been hypothesized, "the impact of MeCP2 malfunction on global DNA methylation and its large-scale effect in RTT has been poorly investigated and is still elusive." Initial reports of compensatory DNA methylation differences between Rett patients and controls in small cohorts were *not confirmed* in larger cohorts. The methylation substrate appears largely preserved.

By contrast, transcriptional alterations in Rett are well-documented: thousands of differentially expressed genes between MECP2-mutant and wildtype neurons, with characteristic patterns affecting long genes [61], synaptic plasticity genes, and activity-dependent genes. The emergent transcriptional operator is substantially disrupted while the substrate is largely preserved — exactly the framework's prediction.

8.3 Generalization to other reader-writer disorders

The framework predicts that similar bioinformatic patterns hold across the reader-writer disorder class:

Fragile X syndrome (FMR1). FMR1 encodes an RNA-binding protein that regulates translation of hundreds of target mRNAs. Loss of FMR1 should leave the RNA target landscape intact (substrate preserved) but disrupt translation regulation (emergent operator). The empirical pattern matches: FMR1-null cells show extensive translational dysregulation while transcriptome changes are more modest [62, 63]. The "substrate" in this case is the RNA target repertoire and ribosome state; FMR1 is the reader.

Angelman syndrome (UBE3A). UBE3A is an E3 ubiquitin ligase functioning as a writer of degradation marks (ubiquitin tags) on target proteins. The protein substrate landscape is preserved; what is disrupted is the writing of degradation marks. The framework predicts protein-level dysregulation (accumulation of UBE3A targets) with relatively preserved transcriptome and methylome. Empirical confirmation: proteomic studies in Angelman models show selective accumulation of UBE3A substrates [64], with transcriptomic changes secondary to the proteomic dysregulation.

Kabuki syndrome (KMT2D). KMT2D is a histone H3K4 methyltransferase

— a writer of activating histone marks at enhancers. Loss of KMT2D should leave the chromatin substrate (nucleosome positions, DNA methylation) intact but disrupt the deposition of H3K4me1/me2/me3 marks. The framework predicts altered H3K4 methylation patterns at affected enhancers with relatively preserved DNA methylation and broader chromatin features. Empirical pattern: ChIP-seq studies of KMT2D mutants show specific loss of H3K4 marks at enhancers, with other chromatin features less affected [65].

ATR-X syndrome (ATRX). ATRX is a chromatin remodeler that maintains heterochromatin at telomeres, centromeres, and tandem repeat sequences. Loss of ATRX should disrupt heterochromatin maintenance at these sites while leaving euchromatin and most of the methylome intact. The framework predicts site-specific chromatin defects with broad methylome preservation. Empirical pattern: ATRX-deficient cells show specific defects at heterochromatin maintenance sites with preservation of broader chromatin architecture [66].

The common pattern across the class: the substratum that the disrupted reader/writer/eraser acts on is preserved; the emergent operator at the level affected by the machinery is disrupted; other levels are relatively preserved.

8.4 Differential diagnosis through multi-omic bioinformatics

The framework's predicted patterns provide a bioinformatic differential diagnostic. A patient with a developmental disorder of unknown cause can be assessed for reader-writer-disorder pattern through paired multi-omic profiling:

1. **DNA methylation** (whole-genome bisulfite sequencing or EPIC array). Preserved methylation argues against substrate-disruption disorder; altered methylation argues for it.
2. **Histone modifications** at affected loci (ChIP-seq for H3K4me3, H3K27me3, H3K27ac). Loss of specific marks argues for writer disruption (KMT2D, EZH2, etc.); preservation argues against.
3. **Chromatin accessibility** (ATAC-seq). Altered accessibility at specific elements with preserved bulk accessibility argues for reader/writer disruption; bulk accessibility changes argue for global disruption.
4. **Transcriptome** (scRNA-seq). Characteristic dysregulation patterns (long genes for Rett; ribosomal targets for Fragile X; UBE3A target accumulation for Angelman) provide signature for specific disorder.

The framework predicts that multi-omic patterns can distinguish reader-writer disorders from substrate-affecting disorders even before the causal mutation is identified — a useful diagnostic capability for patients with unclassified developmental syndromes.

8.5 Therapeutic implications and bioinformatic monitoring

The Medicine paper [Medicine §10.7] establishes that reader-writer disorders are reversibility-prone because the substrate has been preserved. Gene therapy (AAV-based MECP2 replacement, UBE3A reactivation) and pharmacological replacement (trofinetide for Rett, FDA-approved 2023) target this property. Bioinformatic monitoring of therapy outcomes should follow the framework's structural picture:

(i) Therapeutic restoration of reader/writer function should produce *transcriptome normalization* if the substrate is intact, with kinetics determined by the substrate's stability and the cell's ability to read the restored signal. Methylation patterns should remain stable through treatment.

(ii) Patients who do not respond to reader/writer restoration may have suffered substrate degradation in addition to machinery loss — a more advanced disease state. Methylation profiling at baseline could predict therapeutic response: patients with preserved methylation should be more responsive than patients with altered methylation.

(iii) Long-term monitoring through multi-omic profiling should track substrate fidelity (methylation stability) alongside emergent operator function (transcriptome, behavior), with the framework predicting that substrate integrity is the prognostic marker for sustained response.

8.6 Practical consequences

Three framework-grounded predictions for reader-writer disorder bioinformatics:

(P20) Reader-writer disorders should show characteristic multi-omic patterns: substrate preservation (methylation, sequence) with selective emergent operator disruption (transcriptome, chromatin, proteome at the level of the disrupted machinery). The pattern is testable across the class through standardized multi-omic profiling.

(P21) Differential diagnosis between reader-writer disorders and substrate-affecting genetic disorders is achievable through multi-omic patterns even without sequencing the causal gene. This enables diagnostic workup for patients with unclassified developmental syndromes.

(P22) Therapeutic response to reader/writer restoration therapy is predicted by baseline substrate integrity. Methylation profiling at treatment initiation should predict response duration and degree.

The second and third predictions have direct clinical applications and motivate prospective multi-omic studies in reader-writer disorder cohorts.

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9. Synthesis

9.1 Five domains, one structural explanation

The preceding sections have developed framework predictions across five active methodological domains in single-cell bioinformatics. The findings are summarized in Table 1.

Domain	Documented failure mode	Framework's structural explanation	Predicted improvement	Empirical confirmation
Trajectory inference (§2)	scVelo trajectory reversal on complex datasets [3]	Chromatin hidden sector violates Markovian assumption	Multi-omic velocity	MultiVelo [17], mmVelo [18], MoFlow [19] confirm
GRN inference (§3)	Causal methods fail to outperform correlation [5]	Marginal distribution insufficient for causal recovery	Multi-modal methods + prior knowledge	SCENIC+ [23], LINGER [24] confirm
Perturbation prediction (§4)	Deep learning underperforms linear baselines [6]	Information-theoretic ceiling on transcriptome-only data	Knowledge graphs, multi-modal data	GEARS [32], PerturbNet [35] partially confirm
Multi-omics integration (§5)	Static factor analysis misses temporal hierarchy [8]	Modalities measure different memory layers	Timescale-aware integration	QuRIE-seq [8], pseudotemporal methods [41] emerging
Cancer methylome (§6)	(No failure; methylation classifies 95% accurately [43])	Substratum preserved, emergent dysregulated in cancer	Substratum-preservation as diagnostic basis	Methylation classification paradigm operationalizes

Each row instantiates the same underlying structural picture: visible-sector data is insufficient for the task because cellular dynamics depends on hidden-sector memory; methods that include hidden-sector information explicitly succeed; methods that try to extract hidden-sector information from visible-sector data alone are bounded by an information-theoretic ceiling.

9.2 The information-theoretic ceiling

The framework's most consequential bioinformatic prediction is the existence of an information-theoretic ceiling on transcriptome-only methods. This is a strong claim with specific empirical content: no method using only scRNA-seq data — regardless of model architecture, training data size, or computational scale — can recover certain quantities that depend on hidden-sector variables.

The quantities subject to the ceiling include: the cell's causal regulatory network structure (§3), the response to unseen perturbations (§4), the temporal lag relationships between chromatin and transcription (§5), and the substratum-level memory operator (§6). For each, the underlying biology integrates over hidden-sector variables before the data is collected; the integration is irreversible from the visible-sector data alone.

The ceiling is empirically demonstrated in the foundation-model literature [6, 7]. Multiple foundation models trained on tens of millions of cells with billions of parameters fail to outperform simple baselines on tasks subject to the ceiling. This is the empirical signature of a structural limit, not a methodological gap to be closed by better algorithms.

The practical implication is direct: methodological development that stays within the transcriptome-only paradigm faces diminishing returns. Continued scaling of foundation models will produce marginal improvements in tasks where the ceiling has not yet been reached (cell type annotation in well-characterized tissues, gene expression imputation), but cannot produce qualitative improvements in tasks subject to the ceiling (causal GRN inference, perturbation prediction for unseen drugs). The research investment justification for continuing to scale foundation models depends on the assumption that more data and compute will eventually overcome the apparent limits; the framework predicts this assumption is wrong for ceiling-bounded tasks.

9.3 The roadmap forward

The framework's positive prediction is that methodological progress on ceiling-bounded tasks requires architectural changes that incorporate hidden-sector information. Three pathways are consistent with the empirical pattern:

Pathway 1: Multi-modal data. Paired measurements of transcriptome with chromatin accessibility (10x Multiome, scNMT-seq), histone modifications (CITE-seq variants, Paired-Tag), methylation (scNMT-seq), or chromatin architecture (scHi-C) provide direct measurement of hidden-sector variables. Methods using paired data have empirically demonstrated improvements across all five domains (MultiVelo, SCENIC+, multi-modal perturbation methods, timescale-aware integration, multi-omic classifiers).

Pathway 2: Prior biological knowledge. Knowledge graphs (Gene Ontology, KEGG pathways, protein-protein interaction networks), motif libraries

(JASPAR, HOCOMOCO), and curated regulatory databases (Regulome, ENCODE TFBS) encode hidden-sector structure that has been determined through other experimental approaches. Methods incorporating this prior knowledge as structural constraints (GEARS, SCENIC, LINGER) outperform purely data-driven methods.

Pathway 3: Explicit memory state representation. Models that maintain explicit chromatin-state variables alongside transcriptome variables — separate from the latent representations of deep learning models — can in principle capture non-Markovian dynamics. This approach is less developed but theoretically aligned with the framework's prediction. Mechanistic models in the MultiVelo style (coupled ODEs for chromatin and transcription) are early examples.

The three pathways are complementary, not competing. The empirical pattern suggests that combining them — multi-modal data analyzed with knowledge-graph priors using methods that maintain explicit memory state — represents the most promising methodological direction.

9.4 The substratum-emergent operator distinction as the unifying structural feature

A more compact summary of the synthesis: cellular bioinformatics operates on systems that exhibit the substratum-emergent operator distinction [SM §8.7, Medicine §9.4]. The substratum operators — chromatin state, DNA methylation, protein abundance, cell architecture — carry slow memory with high information content. The emergent operators — transcriptome state, instantaneous cell behavior — carry fast information with limited memory. Methods that read the substratum operators (methylation classifiers, multi-omic methods) work; methods that try to infer everything from emergent operators alone (RNA velocity, transcriptome-only foundation models) fail in characteristic patterns.

This is the same structural pattern that organizes the SM's treatment of strong CP, baryon number conservation, sphalerons, and 't Hooft anomaly matching [SM §8.7], and that organizes Rett syndrome paradigm and clinical epigenetic therapeutics [Medicine §9.4]. The pattern is unified across foundational physics, cellular biology, and computational biology — instances of the same structural feature of trace-out dynamics in C1–C4 systems.

The bioinformatic instantiation of the pattern has the empirical advantage of being directly measurable: the substratum is accessible through methylation arrays, ChIP-seq, and ATAC-seq; the emergent operators are accessible through transcriptomics; their dissociation is documentable in patient samples and tissue measurements. This is the strongest empirical confirmation of the framework's structural pattern across all its applications: the bioinformatics literature has independently developed methods that operationalize the substratum-emergent distinction without explicit reference to the underlying framework, and the methods that work are precisely the ones the framework predicts will work.

10. Open Questions and Limits

10.1 What the framework predicts but has not been tested

Several framework predictions in this paper are testable but not yet empirically verified:

(i) **Reader/writer/eraser variant pathogenicity stratification** (§7). The prediction that AlphaMissense and similar methods systematically miscalibrate machinery-gene variants is testable through reanalysis of existing predictions. The result, positive or negative, would be informative.

(ii) **Compositional generalization and chromatin co-localization** (§4.5). The prediction that double-perturbation prediction difficulty correlates with the chromatin co-localization of the perturbed genes can be tested using existing Perturb-seq datasets stratified by chromatin features.

(iii) **Unclassifiable tumor substratum disruption** (§6.4). The prediction that unclassifiable CNS tumors have specifically altered methylation at maintenance sites versus content sites is testable in the existing Heidelberg dataset.

(iv) **Timescale-aware integration superiority** (§5.6). The prediction that timescale-aware multi-omics methods systematically outperform static methods on dynamic biological processes is testable through controlled benchmarking on time-series multi-omic data.

Each is a discrete empirical test with clear positive and negative outcomes. The framework's bioinformatic predictions are falsifiable in a specific operational sense.

10.2 What the framework does not predict

The framework's predictions are limited to information-theoretic constraints and structural patterns. It does not predict:

(i) **Specific algorithm performance.** The framework predicts that multi-modal methods will outperform RNA-only methods, but does not predict which specific multi-modal method will be best for a given task. Algorithm-level performance depends on implementation details, hyperparameter tuning, and dataset-specific features that are outside the framework's scope.

(ii) **Quantitative effect sizes.** The framework predicts that knowledge-graph methods improve over data-driven methods, but does not predict the magnitude of improvement for a specific dataset. Effect sizes depend on the relevance of the available prior knowledge to the specific task.

(iii) **Failure modes outside the ceiling-bounded tasks.** The framework's information-theoretic ceiling applies to tasks requiring hidden-sector information. Other failure modes (batch effects, technical noise, dropouts, low cell numbers) are real and important but outside the framework's prediction scope.

(iv) **Specific molecular mechanisms.** The framework predicts structural patterns at the level of memory layers and information flows, not specific molecular pathways. Mechanistic biology continues to require domain-specific theoretical and experimental work.

10.3 Open structural questions

Three open questions remain at the structural level:

Q1. What is the empirical τ_B hierarchy across biological systems? The framework predicts that different memory layers have different characteristic timescales, with consequences for which modalities matter for which questions (§5.4). The actual τ_B values are tissue- and process-dependent, and have not been systematically characterized. A program of measurements characterizing τ_B across cell types, tissues, and biological processes would substantially refine the framework's bioinformatic predictions.

Q2. How do different machinery genes contribute to the trace-out? The framework treats epigenetic readers, writers, and erasers as components of the trace-out machinery, but the specific contribution of each to the substratum-to-emergent translation is not characterized in detail. Different machinery genes may produce different patterns of substratum-emergent dissociation, with consequences for the differential bioinformatic signatures of different reader-writer disorders (§8.3).

Q3. What is the relationship between the framework's bioinformatic predictions and explicit kinetic models? The framework's predictions are information-theoretic and structural. Specific kinetic models (mass-action models of gene regulation, stochastic gene expression models, polymer models of chromatin) are mechanistic. The relationship between the framework's structural picture and the mechanistic models has not been worked out in detail. A formal connection would clarify which aspects of mechanistic modeling correspond to which framework structural features.

10.4 Limits

The framework's bioinformatic predictions have inherent limits:

(i) **Scope limited to cellular systems satisfying C1–C4.** Systems where C1–C4 fail (rapidly equilibrating systems, low-capacity hidden sectors, decoupled visible-hidden dynamics) do not exhibit the framework's predicted patterns. Not every biological system has substantial τ_B at relevant timescales; the framework's predictions apply where C1–C4 hold.

(ii) **Information-theoretic ceilings are upper bounds, not necessarily attainable.** A method may underperform the ceiling for many reasons unrelated to the framework: poor implementation, insufficient training data, wrong loss function. The framework predicts a ceiling but does not guarantee that any method will reach it.

(iii) **Empirical validation requires controlled benchmarking.** The framework's predictions are about structural patterns in method performance, which requires careful experimental design to test. Confounders (batch effects, dataset selection, evaluation metric choice) can obscure the patterns the framework predicts.

(iv) **The framework does not replace mechanistic biology.** Bioinformatic methods aim to extract biological insight from data, and biological insight requires understanding mechanisms. The framework provides a structural perspective on what mechanisms determine method performance, but does not substitute for the mechanistic understanding itself.

These limits define the framework's scope of application in computational biology. Within the scope — systems with substantial hidden-sector memory analyzed by methods that may or may not include hidden-sector information — the framework's predictions have empirical content that has been confirmed across multiple methodological domains.

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